

The Cognitive Agent

Prospect-theoretic perceived risk. Stateful. Four components.

1. The four components

All measurable at $t - 1$. Only shock is symmetric in sign.

Loss — gains contribute nothing:

$$L_t = \lambda \cdot \max(-r_{t-1}, 0) \quad (1)$$

Shock — deviation from recent norm; surprise, not pain:

$$S_t = \kappa \left| r_{t-1} - \frac{1}{h} \sum_{i=1}^h r_{t-i} \right| \quad (2)$$

Drawdown — reference dependence against a ratcheting peak:

$$P_t = \max(P_{t-1}, V_{t-1}) \quad (3)$$

$$D_t = \delta \cdot \frac{P_t - V_{t-1}}{P_t} \quad (4)$$

Memory — geometric accumulation of past losses:

$$m_t = \rho m_{t-1} + \max(-r_{t-1}, 0), \quad m_0 = 0 \quad (5)$$

$$M_t = \mu m_t \quad (6)$$

Unrolled, every loss the agent ever sustained remains present:

$$m_t = \sum_{k=1}^t \rho^{k-1} \max(-r_{t-k}, 0) \quad (7)$$

2. Aggregation and decision

$$R_t = L_t + S_t + D_t + M_t \quad (8)$$

$$w_t = \text{clip}\left(\frac{1}{1 + sR_t}, w_{\min}, w_{\max}\right) \quad (9)$$

$$V_t = V_{t-1}(1 + w_t r_t) \quad (10)$$

3. Variables

Symbol	Meaning	Value
r_t	Total return on day t	data
L_t, S_t, D_t, M_t	Loss, shock, drawdown, memory components	computed
m_t	Raw memory accumulator (state)	computed
P_t	Running peak of V (state , nondecreasing)	$P_0 = 100$
R_t	Perceived risk — the central quantity	computed
w_t	Exposure for day t	computed
V_t	Portfolio value after day t	$V_0 = 100$
λ	Loss aversion	1.50
κ	Shock sensitivity	1.20
δ	Drawdown regret	1.00
ρ	Memory decay per day	0.92
μ	Memory scale	0.42
s	Risk sensitivity	8.00
h	Shock reference window (days)	10
$w_{\min}, w_{\max}$	Exposure bounds	0.05, 1.00

4. Welfare metrics

Let $q_{0.9}$ be the empirical 90th percentile of $\{R_t\}_{t=1}^T$ and $\mathcal{S} = \{t : R_t > q_{0.9}\}$.

$$\text{Baseline stress} = \bar{R} = \frac{1}{T} \sum_t R_t \quad (11)$$

$$\text{Stress volatility} = \left[\frac{1}{T} \sum_t (R_t - \bar{R})^2 \right]^{1/2} \quad (12)$$

$$\text{Peak stress} = \max_t R_t \quad (13)$$

$$\text{Time in stress} = |\mathcal{S}|/T \equiv 0.10 \quad (\text{by construction}) \quad (14)$$

$$\text{Mean episode length} = |\mathcal{S}|/\#\{\text{maximal runs in } \mathcal{S}\} \quad (15)$$

5. Properties

Stateful. m_t and P_t carry across days. At $\rho = 0.92$ a loss retains half its weight after $\ln 2 / \ln(1/\rho) \approx 8.3$ trading days.

Path-dependent. Two histories with identical trailing 20-day windows give identical *classical* exposures but may give very different cognitive ones.

Bounded and convex. $R \mapsto (1 + sR)^{-1}$ equals 1 at $R = 0$, is strictly decreasing and convex, and tends to 0. The agent can be frightened but never fully exits. Convexity matters wherever exposures are averaged.

Asymmetric except in shock. L_t and M_t ignore gains entirely; D_t responds only below peak. S_t alone is symmetric.